

Ella Potter

SOFTWARE ENGINEER | FULL STACK | MOBILE | EMBEDDED SYSTEMS

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EDUCATION

University of Iowa | Bachelor of Science in Engineering

December 2026

Major: Computer Science Engineering

Relevant coursework: *Data Structures, Algorithms, Operating Systems, Embedded Systems, Software Engineering, Artificial Intelligence, Database Systems, etc.*

TECHNICAL SKILLS

Languages: Python, Java, C++, Dart, Ruby, JavaScript, SQL, Embedded C, AVR Assembly, Verilog

Frameworks: Flutter, Flask, Ruby on Rails, Jinja, HTML/CSS, RSpec, Cucumber, pytest

Data and Tools: MySQL, PostgreSQL, SQLite, Git, GitHub, Docker, Arduino, Stripe, OpenAI API

SOFTWARE ENGINEERING EXPERIENCE

Software Engineering Intern | RideMatch | Iowa City, IA

May 2026 - Present

- Advance a three-client ridesharing prototype toward release by improving Flutter rider and driver applications and a Flask administrative portal.
- Integrate and debug mobile interfaces, Python services, MySQL data, and payment workflows; support automated testing, deployment, and cross-client scenarios.
- Partner with the project sponsor to translate business needs into technical features while coordinating team tasks, documentation, and GitHub workflow.

Lead Coding and Robotics Instructor | iCode | Naperville, IL

May 2025 - Aug 2025

- Led coding, robotics, and gaming lessons for students ages 6-15 using Python, Java, and C#
- Coached students through debugging and problem solving.

Additional Work Experience

2020 - Present

Worked consistently since age 16 in social media, guest services, food service, youth programs, and retail while completing high school and college.

HIGHLIGHTED ENGINEERING PROJECTS

GiftWise | *Ruby on Rails, RSpec, Cucumber, OpenAI API, OAuth*

Built an AI-powered gift recommendation web app that uses recipient profiles, relationships, occasions, and budgets; implemented account features, Google OAuth, validation, and automated tests.

Library Database Management System | *Python, Flask, MySQL, JavaScript*

Built a library operations web app for patron and staff checkout, reservation, fee, and inventory workflows; designed a 23-table MySQL database with five triggers, views, a stored function, and a stored procedure.

Election Management System | *Python, Flask, MySQL, pytest*

Created a secure voting web app with voter, election manager, and administrator roles; built account verification, precinct-specific ballots, protected sessions, result controls, and automated tests on a 21-table schema.

Embedded Digital Compass | *ATmega328P, Embedded C, AVR Assembly, I2C*

Built a microcontroller-based navigation system that reads an MMC3416xPJ magnetometer over I2C and drives an LCD and stepper compass needle; debugged mixed-voltage hardware and firmware with lab instruments.

LEADERSHIP

Society of Women Engineers | University of Iowa **Youth Group Leader** | Hope City Church